

Intelligent Humanoid Airport Concierge: A RAG-Based Voice Interface and Autonomous Navigation System for the Unitree G1

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I. INTRODUCTION

Airport terminals need both natural-language guidance and physical wayfinding, a combination existing systems only partly cover: chatbots answer questions but cannot escort a passenger, while navigation robots move reliably but understand little natural language. This work integrates both on a Unitree G1 EDU humanoid. A retrieval-augmented voice interface answers passenger queries from an approved knowledge base, and a LiDAR-based navigation stack then guides the passenger to the identified destination. The system targets English, Sinhala and Tamil and is validated in MuJoCo/ROS 2 simulation.

II. LITERATURE REVIEW

Retrieval-augmented generation (RAG) grounds language-model outputs in an external document store, reducing hallucination and allowing knowledge updates without retraining [1]. The ROS 2 Nav2 stack offers a modular, behaviour-tree-driven planner and controller demonstrated on service robots operating alongside pedestrians [2]. Multilingual transcription remains a bottleneck for lower-resource languages such as Sinhala and Tamil; Whisper generalises well but can trail cloud recognisers on noisy, accented speech [3], and faster-whisper [5] reimplements it to cut inference time and memory. Few prior systems combine RAG conversation, multilingual speech and full-body humanoid navigation in one airport-concierge pipeline.

III. MATERIALS AND METHODS

A. Problem Formulation and Objectives

The target scenario is a passenger asking, for example, “Where is the taxi pickup point?” The robot must transcribe

the request, retrieve approved information, give a grounded spoken answer, identify the destination on its map, and guide the passenger there autonomously. The objectives are to: (1) build a voice interface for the G1; (2) ground answers in approved documents; (3) support English, Sinhala and Tamil; (4) map, localize and navigate to a selected destination, validated in simulation first; and (5) evaluate accuracy, latency, navigation success and computational cost.

B. System Architecture

Fig. 1 shows the two pipelines. In the conversation pipeline, WebRTC voice-activity detection (VAD) segments speech, which is transcribed and passed to a LangChain agent. The agent retrieves context from a FAISS index of approved documents (Gemini embeddings) and a Gemini model generates a grounded spoken answer. If the request names a destination, the agent forwards it as a Nav2 goal, planned on a LiDAR-built map, localized through the frame chain odom → pelvis → mid360_link, and executed as velocity commands on the simulated G1.

C. Conversational RAG Pipeline

Table I lists the components chosen for the conversation pipeline; speech recognition uses Whisper, with faster-whisper [5] as a lighter local backend. RAG was preferred over fine-tuning because the knowledge base can be updated by re-indexing documents without retraining, which matters where airport signage, routes and services change.

D. Localization and Navigation

Indoor maps are built in MuJoCo/ROS 2 simulation from a simulated Livox MID-360 LiDAR, an Intel RealSense D435 depth camera, an IMU and ground-truth odometry

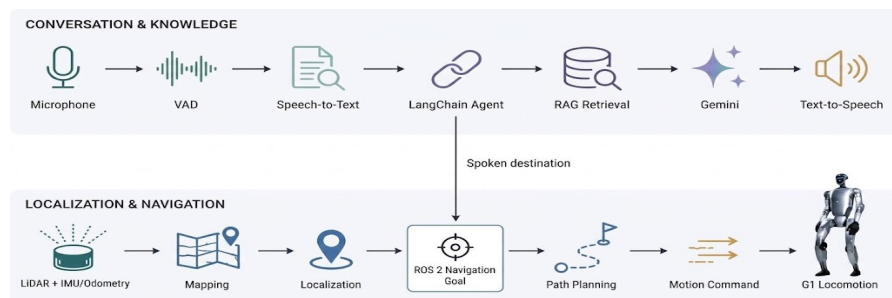


Fig. 1. System architecture: the conversation pipeline (top) passes a spoken destination to the ROS 2 localization and navigation pipeline (bottom).

(Fig. 2). Given a 2D map and an initial pose, Nav2 plans a global path, converts it into local velocity commands and continuously updates the pose and obstacle estimate. To keep the humanoid stable, velocity limits are set conservatively at about 0.2 m/s linear and 0.3 rad/s angular.

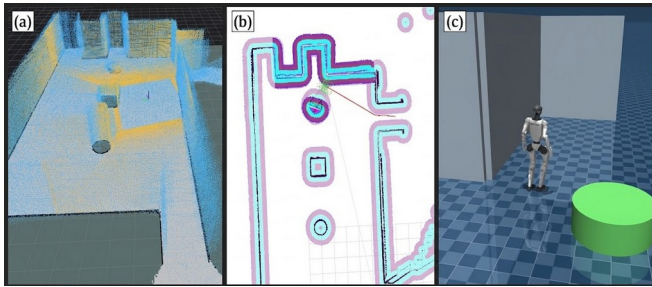


Fig. 2. (a) 3D point cloud visualization in RViz[4]; (b) Nav2 costmap and LiDAR scan in RViz; (c) Unitree G1 Hmanoid robot walking in the MuJoCo simulation.

TABLE I. METHOD SELECTION AND JUSTIFICATION

Component	Method	Justification
Speech detection	WebRTC VAD	Lightweight; avoids processing silence
Speech recognition	Whisper / faster-whisper; cloud STT	Multilingual (EN/SI/TA) with a local backup
Retrieval and grounding	Gemini embeddings + FAISS; RAG with Gemini	Efficient search; restricts answers to approved sources
Orchestration	LangChain	Connects conversation, retrieval and navigation
Mapping / navigation	LiDAR SLAM + Nav2	Mature, modular indoor navigation framework

IV. RESULTS AND DISCUSSION

The G1 MuJoCo simulation, simulated LiDAR and depth sensing, ROS 2 odometry/TF, voxel mapping, RViz visualization and text-to-speech output are complete. Autonomous navigation to a selected goal and English RAG conversation have been demonstrated end-to-end, with grounded English answers and reliable mapping and goal navigation in simulation. Sinhala and Tamil recognition is still being tuned: local Whisper models are less reliable for these languages, and a cloud recogniser improves them. Running perception and simulation together slows navigation, and 3D obstacle avoidance with the depth camera is not yet implemented, so the 3D LiDAR may miss certain types of obstacles.

Conversational grounding quality was measured with the RAGAS framework [6], which scores each question-answer pair on four dimensions using an LLM judge: Faithfulness (whether the generated answer's claims are supported by the retrieved context), Answer Relevancy (whether the answer addresses the question asked), Context Precision (whether relevant retrieved chunks are ranked near the top) and Context Recall (whether the retrieved context contains the information needed to answer). Both the RAG generator and the RAGAS judge used Gemini 3.5 Flash-Lite, with $k = 3$ chunks retrieved per query and answer-relevancy strictness set to 1 due to free-tier API limits. Fig. 3 reports the mean score per dimension over a held-out set of $n = 16$ English

test questions spanning airport wayfinding, services and policy queries, with error bars showing one standard deviation across questions.

All 16 questions completed with zero evaluation errors. The pipeline achieved a mean Faithfulness of 0.8979 (± 0.1991), Answer Relevancy of 0.8837 (± 0.1280), Context Precision of 0.8906 (± 0.2167) and Context Recall of 0.9792 (± 0.0833), for a macro-average RAGAS score of 0.9129. Context Recall was both the highest and most consistent score, indicating the FAISS retriever usually captured the information needed for the reference answers. The main weaknesses were on capability-related questions and retrieval ranking for some location questions, which pulled down Faithfulness and Context Precision respectively; these are the main targets for improving answer quality. As RAGAS scores are LLM-judge estimates rather than ground-truth accuracy, they should be read as reference-free proxies for answer quality pending correlation with human-annotated judgments.

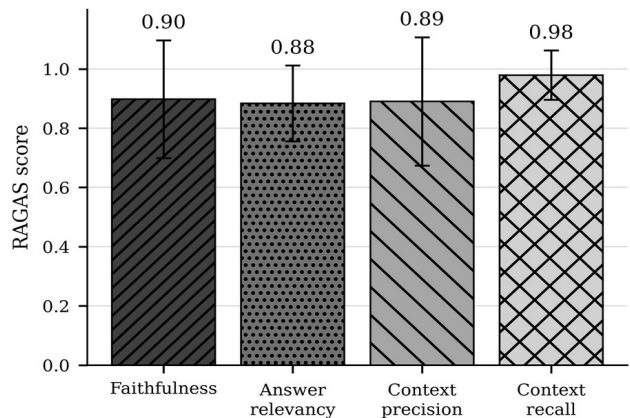


Fig. 3. Mean RAGAS evaluation scores (Faithfulness, Answer Relevancy, Context Precision, Context Recall) over $n = 16$ test questions, scored by an LLM judge (Gemini 3.5 Flash-Lite). Error bars show one standard deviation across questions.

V. CONCLUSION AND FUTURE WORK

A single humanoid platform can combine grounded conversational AI with autonomous indoor navigation for an airport-concierge role, with English conversation and simulated goal navigation working end-to-end. Future work will link spoken destinations to Nav2 goals, replace ground-truth odometry with LiDAR-inertial localization, improve Sinhala and Tamil recognition, add depth-based 3D obstacle detection, and move to the physical G1 for tests with real users.

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